Structural Self-Energy Expansion (SSEE): A Minimal-Parameter Dark Energy Framework — Algebraic Derivation, Boltzmann Validation, and Effective Field Theory Stability Analysis

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November 2026

Abstract

We present the Structural Self-Energy Expansion (SSEE), a dark energy framework that derives its cosmological background sector algebraically from the golden ratio φ and π, with zero fitted dimensionless parameters at the background level and no WIMP, QCD-axion, or SUSY dark-matter particle. Phenomenological extensions in the dark-matter sector, proposed in an earlier version of Paper 6, are *retracted* (§5.2): the matter sector is a single unpartitioned component, and their register entries (OP-9, OP-11) close by dissolution (`OPEN_PROBLEMS.md`). In CMB-likelihood terms SSEE is a $k = 2$ fit: of ΛCDM's six parameters it fixes four algebraically (Ω_b, Ω_c via $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$, $n_s = 1 - \varphi^{-7}$, and the global rate $H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})$ from the measured SH0ES input, which lands on the pure number $3(\varphi + \pi)^2$), leaving only the primordial amplitude A_s and the optical depth τ ($k = 2$ versus $k = 6$, the count used in the Paper 3 model comparison). The dark energy equation of state $w_0 = -(3\varphi + \pi)/[2(\varphi + \pi)] = -0.839950$ and $w_a = -0.669975$ are fixed by the algebraic construction and agree with the official DESI DR2 w_0w_aCDM posterior within 0.24σ (Pantheon+; 0.2–1.8σ across SN compilations)—a parameter-free postdiction, with no claim of temporal priority over DESI DR2. A Bayesian MCMC analysis under the global rate H_0^{glob} as prior ($H_0^{\mathrm{prior,SSEE}} = H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}}) = 67.962 \pm 0.968$ km s^{-1} Mpc^{-1}, which lies 0.90σ from the value at which Planck `plik_lite` is minimised with SSEE-fixed ω_b, ω_c; Papers 3, 9–10) yields $H_0 = 67.82^{+0.41}_{-0.41}$ km s^{-1} Mpc^{-1} (0.68σ from Planck, 0.34σ from H_0^{glob}). Model selection favours SSEE by $\Delta\mathrm{BIC} = -10.90$ over ΛCDM and -9.32 over CPL with nothing free ($k = 0$), and -5.81 and -4.24 in the conservative version that samples H_0 and $\Omega_b h^2$ ($k = 2$). Probe by probe, with the background of both models fixed in advance, SSEE matches ΛCDM within $|\Delta\chi^2| \leq 3.0$ in 10 of 11 comparisons—the supernova compilations, ACT and SPT CMB lensing,

DES Y3, Planck and KiDS-Legacy—even where ΛCDM fits up to 6 parameters against SSEE's $\{A_s, \tau\}$, and BOSS full-shape gives $\Delta\chi^2 = +2.84$ with SSEE fully fixed against ΛCDM with its cosmology free; DESI DR2 BAO is the only probe that separates the two models, by $\Delta\chi^2 = -20.6$ in favour of SSEE. A complementary full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` + lensing + DESI DR2 + $f\sigma_8$, w_0, w_a free and H_0 under a flat prior) places the model at 1.31σ in the w_0–w_a plane, where ΛCDM is excluded at 3.22σ; under this uncompressed CMB likelihood H_0 relaxes to 65.87 ± 1.17 (1.79σ below the global rate).

The CMB-sector matter density is the standard physical observable $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ (Planck: 0.3153±0.0073), built piece-by-piece from SSEE algebra (ω_m-direct; OP-8 dissolved, no matter factor) and distinct from the saturation $s_m = 1 + w_0 = 0.160050$, which fixes the DESI BAO equation of state and is not a density at all. With this single density, full-sky CLASS Boltzmann runs place the acoustic peaks where ΛCDM does (RMS residual 0.18%).

A linear perturbation analysis with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the Poisson equation, with the dark-energy sector stabilised by the sound speed of the field of Paper 7, yields a growth factor $\mathcal{G} = 1.0032$ (Paper 5) and, with A_s held at Planck's value, $\sigma_8 = 0.8136$, giving a mean $f\sigma_8$ tension of 0.70σ across six RSD surveys — statistically identical to ΛCDM (0.73σ). In the lensing amplitude, against the 357 *raw* KiDS-Legacy $\xi_\pm$ points, with a *single* unpartitioned matter sector, the background fixed by the algebra and A_s fixed to the value the model's own CMB fit supplies — *no* free cosmological parameter — the model predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 measured (0.04σ).[1] **There is no S_8 tension** (§5.2).

At the Bellini-Sawicki level the sector is minimal Horndeski, $\alpha_T = \alpha_M = \alpha_B = 0$ (GW170817 compatible), with kineticity $\alpha_K = 15.591336$ and a ghost-free, gradient-stable sound speed $c_s^2 = 0.021284$ (Paper 7).

We additionally summarise three theoretical extensions of the framework. In the strong-field regime Paper 8 keeps a disformal contrast, in which the photon deflection would be amplified by $\sqrt{\mathrm{AURA}} = 1.9995$ (numerically $1.00027\,\mathcal{M}$ but algebraically distinct) and which makes no prediction of the model, and derives the canonical limit with EFT $\alpha_B = \alpha_M = \alpha_T = 0$, in which the residual fifth force is suppressed to $F_\phi/F_N \sim 6 \times 10^{-13}$ at 10 kpc and the observable lensing is that of GR with the model's cold matter; current SLACS/BELLS data are consistent with it (Paper 8; OPEN_PROBLEMS.md OP-13). A late-time local screening fraction $f_{\mathrm{scr}}^{\mathrm{IR}} = s_K/(3\mathcal{M}) = (\pi - \varphi)/(\varphi + \pi)^2 = 0.0673$ relates, under the screening postulate of Paper 9, the locally measured rate to the global background. The measured datum is the input and the global rate is the output: $H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{scr}}^{\mathrm{IR}}) = 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, which sits 0.17σ from the pure number $3(\varphi + \pi)^2 = 67.96214$; with the full infrared-plus-ultraviolet fraction $f_{\mathrm{scr}}^{\mathrm{full}} = 0.069522$ the same operation returns 67.962142, a residual of $+4.2 \times 10^{-6}$ km/s/Mpc against that number, far inside the propagated ±0.968, in both cases with zero free parameters (Papers 9 and 10). A UV completion $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$ whose cutoff $M = 9.68$ meV is conjectured to be set by the inflationary α-attractor (Conjecture C.1, Paper 10) is consistent with the cascade, which does not measure it. All model parameters

[1] Updated 2026-09-26. KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441) was published sixteen months before this analysis; no temporal priority is claimed. Its recalibration of the source redshift distributions dissolves the Planck–shear disagreement in the data themselves: re-running the KiDS-1000 analysis under blinding, the collaboration's Appendix I finds $2.39\sigma \to 1.42\sigma$ from the $n(z)$ calibration alone. With A_s left free, the amplitude the shear asks for lies -0.25σ from the one the CMB fixes under the *same* algebraic background. The three χ^2 compared span 0.74 over 357 points, so the shear does *not* discriminate between the models; what separates them is the parameter count.

are falsifiable by named forthcoming experiments.

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1 Introduction

The DESI DR2 baryon acoustic oscillation data [Abdul-Karim et al., 2025] provide evidence for dynamical dark energy at 2.8–4.2σ depending on the supernova sample, with the DESI + CMB + Pantheon+ posterior centred at $(w_0, w_a) = (-0.838, -0.62)$. This motivates systematic testing of models that predict the dark energy equation of state from theoretical principles rather than fitting it.

The SSEEframework proposes that the macroscopic universe obeys an algebraic closure condition: the dimensionless background parameters are fixed by two dimensionless geometric constants, the golden ratio $\varphi = (1 + \sqrt{5})/2 \simeq 1.6180$ and $\pi \simeq 3.1416$. No optimisation over formula space was performed, with one declared exception: ω_b, whose form was chosen among 7 (φ, π) expressions compared with Planck (Paper 4; OP-1): the parameter values are fixed by the algebraic construction [Almeida Vallejo, 2026k], not adjusted to the data. We make no claim of temporal priority over DESI DR2 (public from 2025 March); the agreement reported below is a parameter-free *postdiction*, and the only timestamped predictions are those reserved for unreleased data (DESI DR3, Euclid).

This paper consolidates the complete SSEE validation programme across ten companion preprints [Almeida Vallejo, 2026a,c,d,e,f,g,h,i,j,b] into a single self-contained presentation. We report four independent validation tiers:

1. **Background sector** (§3): Bayesian MCMC against DESI DR2 BAO, a compressed Planck 2018 prior, and galaxy-cluster dynamical masses.
2. **CMB angular spectrum** (§4): Full-sky CLASS Boltzmann integration with the single ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.308881$), which places the acoustic peaks where ΛCDM does.
3. **Structure growth and σ_8** (§5): Israel-Stewart causal perturbation theory with a single matter sector. Against raw KiDS-Legacy $\xi_\pm$, with A_s fixed by the model's own CMB fit and no free cosmological parameter, it predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176.
4. **EFT stability** (§6): the Paper 7 ghost condensate, gradient-stable with $c_s^2 = 0.021284$, and gravitational-wave compatibility ($\alpha_T = \alpha_M = \alpha_B = 0$).

Section 7 then summarises three theoretical extensions of the framework — the strong-field lensing regime, an algebraic screening account of the Hubble tension (under a postulate), and a UV completion — which are developed in full in the companion Papers 8–10 and are presented here in condensed form.

Throughout, we distinguish three epistemic statuses: *postdiction* (algebraic result fixed by construction, then compared to data already public), *retrodiction* (independently derived, then compared to existing data), and *prediction* (fixed now, tested against not-yet-released data). Table 5 in §8 lists all entries explicitly.

2 Algebraic Framework

2.1 Structural registers

The SSEE framework postulates a closed algebraic system of seven structural registers derived from φ and π:

$$\begin{aligned}
\Omega &= \pi + \varphi \simeq 4.7596 \quad \text{(Stability Metric)} \\
\beta &= (\pi + \varphi)/2 \simeq 2.3798 \quad \text{(Base Coupling Scalar)} \\
\mathrm{KAL}_0 &= \beta + \pi \simeq 5.5214 \quad \text{(Structural Retention)} \\
P_{\mathrm{sc}} &= \Omega + \varphi \simeq 6.3777 \quad \text{(Dynamical Evolution Scalar)} \\
K_v &= \varphi + \pi + \Omega \simeq 9.5193 \quad \text{(Structural Constraint)} \\
T_r &= 3(\varphi + \beta) \simeq 11.9935 \quad \text{(3D Saturation Horizon)} \\
M_v &= \varphi + \pi + K_v \simeq 14.2789 \quad \text{(Maximal Dimensional Invariant)}
\end{aligned} \tag{1}$$

From these seven scalars, the dark energy equation of state is fixed:

$$w_0 = -\frac{T_r}{M_v} = -\frac{3\varphi + \pi}{2(\varphi + \pi)} \simeq -0.83995, \qquad w_a = -\frac{P_{\mathrm{sc}}}{I_g} = -\frac{P_{\mathrm{sc}}}{\pi + P_{\mathrm{sc}}} = -0.669975. \tag{2}$$

The dark-energy saturation follows algebraically: $s_{\mathrm{DE}} = |w_0| = T_r/M_v = 0.839950$ — a number of the equation of state, not the density $\Omega_{\mathrm{DE}} = 1 - \Omega_{m,\mathrm{CMB}} = 0.691119$.

2.2 Derived cosmological parameters

All primary observational inputs are retrodicted without fitting:

$$\frac{H_0^{\mathrm{glob}}}{\mathrm{km\,s^{-1}\,Mpc^{-1}}} = \frac{H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})}{\mathrm{km\,s^{-1}\,Mpc^{-1}}} \simeq 67.96214 \;\; \text{(target } 3(\varphi + \pi)^2\text{)}, \tag{3}$$

$$n_s = 1 - \varphi^{-7} \simeq 0.96556, \tag{4}$$

$$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 \simeq 0.308881 \quad (\omega_m\text{-direct, CMB sector}). \tag{5}$$

Planck 2018 measurements: $H_0 = 67.36 \pm 0.54$, $n_s = 0.9649 \pm 0.0042$, $\Omega_m = 0.3153 \pm 0.0073$ [Planck Collaboration, 2020]. All three SSEE retrodictions lie within 1.5σ of the Planck values with no adjustment.

2.3 One matter density and the CMB

The framework has a single matter density. The dark energy EoS fixes the saturation $s_{\mathrm{DE}} = |w_0| = 0.839950$, and with it $s_m = 1 - s_{\mathrm{DE}} = 1 + w_0 = 0.160050$ — a number of the equation of state, *not* a matter density. However, the CMB angular-diameter distance is sensitive to the *total* matter density, which under the ω_m-direct reframe (OP-8 dissolved) is the standard physical observable built from SSEE algebra:

$$\Omega_{m,\mathrm{CMB}} = \frac{\omega_m}{h^2} = \frac{\omega_b + \omega_c + \omega_\nu}{h^2} = 0.308881, \qquad \omega_c = \mathrm{KAL}_0\, \omega_b\, n_s, \tag{6}$$

in agreement with Planck 2018 (0.3153 ± 0.0073) at 0.88σ. There is *no* matter factor: $s_m = 0.160050$ (a number of the equation of state, tested by DESI through w_0) and ω_m (a density, tested by the CMB) are two independent SSEE predictions, of two different kinds.

(The earlier mapping $\Omega_{m,\mathrm{CMB}} = \mathcal{M} \times s_m$, with $\mathcal{M} = (3\varphi + \pi)/4$, is withdrawn; $\mathcal{M}$ now enters only the Hubble screening fraction, Paper 9.) The physical interpretation of these two independent predictions — the earlier reading as a "two-sector split" is superseded by the ω_m-direct construction — is examined in Papers 3 and 6 [Almeida Vallejo, 2026d,g]. Section 4 checks this density against the acoustic peaks.

3 Background Sector Validation

3.1 Bayesian MCMC setup

We compare SSEE against ΛCDM and a free CPL parametrisation using Bayesian evidence criteria and the EMCEE ensemble sampler [Foreman-Mackey et al., 2013] with $N_{\mathrm{walkers}} = 100$, $N_{\mathrm{steps}} = 25\,000$ (2.5×10^6 total evaluations). The likelihood combines:

- DESI DR2 BAO: 13 data points at $z = 0.295$–2.330 [Abdul-Karim et al., 2025], with the full 13×13 same-bin block-diagonal covariance;
- Planck 2018 compressed Gaussian prior: $H_0 = 67.36 \pm 0.54$, $\Omega_m = 0.3153 \pm 0.0073$, $\Omega_b h^2 = 0.02237 \pm 0.00015$, $\rho(H_0, \Omega_m) = -0.85$;
- as a separate consistency test outside the likelihood, the dynamical masses of all 46 clusters in Tables II–III of Zhang et al. [2026].

For SSEE, (w_0, w_a) are fixed algebraically; H_0 and $\Omega_b h^2$ are sampled in this conservative version ($k = 2$), although ω_b is algebraic and H_0 is the output of the cascade. With nothing free ($k = 0$) SSEE is favoured by $\Delta\mathrm{BIC} = -10.90$ over ΛCDM and -9.32 over CPL, and freeing H_0 and $\Omega_b h^2$ lowers χ^2 by only 0.46 (Paper 2). For ΛCDM, (H_0, Ω_m) are free. For CPL, $(H_0, \Omega_m, w_0, w_a)$ are free.

3.2 Results

Table 1 summarises the key posteriors and information criteria.

Table 1: Bayesian MCMC results (three-model run, DESI DR2 + common Planck prior, total matter density in the background geometry). N_{eff}: effective independent samples. $\Delta \equiv$ SSEE − model; negative Δ favours SSEE.

Quantity	SSEE	ΛCDM	CPL
H_0 [km s^{-1} Mpc^{-1}]	67.87 ± 0.28	68.39 ± 0.28	67.26 ± 0.52
Free parameters k	2	3	5
N_{eff}	79,203	65,893	44,343
ΔBIC	ref	−5.81	−4.24
ΔDIC	ref	−5.02	−1.70
Clusters, χ^2/N (IGIMF, $N = 46$)	67.6 (2.32σ)	68.9 (2.41σ)	—
(w_0, w_a) vs DESI DR2 $w_0 w_a$CDM (Pantheon+)	0.24σ	2.58σ	—

With the total gravitating matter density $\Omega_m = \omega_m/h^2 = 0.308881$ in the background geometry, SSEE is favoured on every information criterion: $\Delta\mathrm{BIC} = -5.81$ over ΛCDM and -4.24 over CPL, $\Delta\mathrm{DIC} = -5.02$ and -1.70. The SSEE sound horizon then matches

ΛCDM ($r_d = 147.17$ vs 147.10 Mpc, CAMB) and the H_0 posterior sits 0.84σ from Planck, so no background-structure penalty arises. An earlier "+206" full-background penalty was an artefact of placing the equation-of-state number $1 + w_0 = 0.160050$ in $E(z)$ where the total density belongs; with the correct total density it does not occur. The single matter density (§2.3) is checked against the acoustic peaks in §4 below.

3.3 Galaxy-cluster mass sector

SSEE does not modify gravity ($\alpha_T = \alpha_M = \alpha_B = 0$, Paper VII), so a cluster weighs its whole matter content: $M_{\rm dyn} = (\omega_m/\omega_b)\, M_{\rm bar} = 6.364\, M_{\rm bar}$, where the cold dark matter enters through the forward identity $\omega_c = \mathrm{KAL}_0\, \omega_b\, n_s$ and no parameter is fitted. Against all 46 systems of Zhang et al. [2026], with baryons that include the stellar remnants of a variable initial mass function (IGIMF), SSEE is consistent at 2.32σ and ΛCDM at 2.41σ; with the canonical IMF both fail (8.64σ and 8.46σ). The two backgrounds differ in $M_{\rm dyn}/M_{\rm bar}$ by under one per cent: clusters test the baryon census and the matter content, not the background (Paper II).

3.4 Multi-probe MCMC validation: two complementary tests

The rigidly fixed SSEE background can be interrogated in two distinct and complementary ways, which must not be conflated. *(i) Free test:* let the data choose the equation of state and ask where they land relative to the algebraic prediction — *"what do the data prefer?"* *(ii) Fixed test:* impose the algebraic values and measure the absolute goodness of fit — *"how well does the imposed model describe the data?"* Both are performed with the *full* CMB power-spectrum likelihood — not a compressed distance prior — so that no ΛCDM-like assumption is smuggled in through the compression.

Free test (what the data prefer). We run an MCMC with CAMB + Planck `plik_lite` TTTEEE + lensing + a τ prior, combined with DESI DR2 BAO and $f\sigma_8$, sampling $(H_0, \Omega_b h^2, \Omega_c h^2, n_s, \tau, w_0, w_a)$ with *flat* priors (H_0 free in $[60, 80]$; four chains, 53,116 samples after a 30% burn-in per chain, Gelman–Rubin $R - 1 = 0.020$). Table 2 lists the recovered posteriors: the tightly constrained CMB combinations $\omega_b, \omega_c, r_{\rm drag}$ are recovered to $\leq 0.2\sigma$ — an independent confirmation of the ω_m-direct reframe from the full C_ℓ spectrum rather than from a compressed prior. The remaining parameters $(H_0, \Omega_m, w_0, w_a)$ sit 1.43–1.79σ from the algebraic values, the uncompressed CMB likelihood pulling H_0 down to 65.87 ± 1.17. In the dark-energy plane the free posterior $(w_0, w_a) = (-0.655, -1.104)$ *excludes* the cosmological constant $(-1, 0)$ at 3.22σ, while the SSEE algebraic point $(-0.840, -0.670)$ lies at only 1.31σ in the correlated ($\rho = -0.93$) w_0–w_a plane — more than twice as close as ΛCDM.[2]

Under the *full* Planck C_ℓ likelihood — with H_0 free under a flat prior rather than fixed to a compressed distance prior — the posterior sits at $H_0 = 65.87 \pm 1.17$, some 1.79σ below the global rate $H_0^{\rm glob} = 67.96214$. This mild downward pull is a known feature of $w_0 w_a$CDM under the uncompressed CMB likelihood and is stated openly. The key model-comparison statement is unchanged and in fact sharper: the fixed algebraic point $(-0.840, -0.670)$ lies 1.31σ from the joint w_0–w_a posterior, whereas the cosmological-constant point $(-1, 0)$ is excluded at 3.22σ. The joint distance is smaller than the indi-

[2]The $\sigma_8 = 0.799$, $S_8 = 0.836$ recovered here comes from this free-(w_0, w_a) fit and is not the model's growth amplitude, which is $S_8 = 0.8273$ (§5.2).

Table 2: Full-likelihood multi-probe MCMC (CAMB + Planck `plik_lite` TTTEEE + lensing + τ prior + DESI DR2 + $f\sigma_8$, with w_0, w_a free and H_0 under a flat prior $[60, 80]$): posteriors vs. algebraic SSEE values. Four independent chains, Gelman–Rubin $R - 1 = 0.020$, 53,116 samples after burn-in. DESI DR2-corrected vector (V-L4-DESI).

Parameter	Posterior (mean $\pm 1\sigma$)	Algebraic SSEE	Tension
$\Omega_b h^2$	0.02243 ± 0.00013	0.022418	0.08σ
$\Omega_c h^2$	0.11935 ± 0.00082	0.119514	0.20σ
$r_{\rm drag}$ [Mpc]	147.20 ± 0.21	147.17	0.10σ
H_0 [km s^{-1} Mpc^{-1}]	65.87 ± 1.17	67.962142	1.79σ
Ω_m	0.3288 ± 0.0118	0.308881	1.69σ
w_0	-0.655 ± 0.107	-0.839950	1.74σ
w_a	-1.104 ± 0.304	-0.669975	1.43σ
Joint w_0–w_a distance: SSEE 1.31σ / ΛCDM excluded 3.22σ			

vidual w_0, w_a marginals (1.74σ, 1.43σ) because the algebraic offset lies *along* the strong w_0–w_a degeneracy ($\rho = -0.93$), not across it. Corner plots and trace diagnostics are archived at Zenodo [Almeida Vallejo, 2026l].

Fixed test (imposing ssee). The complementary question — how well the fixed algebraic model fits the CMB in absolute terms — is answered in Paper III: with $H_0, \omega_b, \omega_c, n_s$ all algebraically fixed and only (A_s, τ) free, the full Planck `plik` likelihood yields $\Delta\text{BIC} = -33.83$ relative to the six-parameter ΛCDM fit (-26.03 for the `plik_lite` point estimate), decisively favouring SSEE on the Jeffreys scale. The advantage is driven by parsimony ($k = 2$ versus $k = 6$), not by a lower χ^2: the fixed model matches ΛCDM spectrum-for-spectrum ($\chi^2_r(\text{TT}) = 1.042$) while spending four fewer parameters. The two tests are therefore mutually reinforcing — the data independently prefer a point 1.31σ from SSEE (with ΛCDM excluded at 3.22σ), and imposing SSEE exactly costs no fit quality while gaining a substantial parsimony advantage.

The same fixed values in every reading. Both tests share a defining feature: the SSEE background carries *the same fixed values into every probe*. The pair $w_0 = -0.839950$, $w_a = -0.669975$ confronts DESI (Paper II, 0.24σ Pantheon+), the Planck 2018 CMB (Paper III, $\Delta\text{BIC} = -33.83$), the $f\sigma_8$ compilation (0.70σ) and the cluster masses (2.32σ) without ever being re-tuned per data set. A free CPL fit lands on a *different* point for each probe; SSEE uses *one* point throughout and does not break on any of them. Fixing the model should have made it fragile — it does not. (A faster cross-check replacing the full likelihood with the Planck compressed prior was used during development as an internal consistency check only; it has no surviving log and no number from it is quoted here.)

3.5 Probe by probe: a tie everywhere, a separation in DESI

The model-selection numbers above weigh parsimony; Table 3 shows the fit itself. Each probe is read with the background of both models fixed in advance—SSEE at its algebraic point, ΛCDM at the Planck 2018 best fit—and, in block B, with ΛCDM fitting parameters that SSEE does not have. SSEE stays within $|\Delta\chi^2| \leq 3.0$ of ΛCDM in 10 of the 11

comparisons, below the $\Delta\chi^2 = 9$ we take as separating, and BOSS full-shape (§5.2) gives $\Delta\chi^2 = +2.84$ with SSEE fully fixed against ΛCDM with its cosmology free. DESI DR2 BAO is the only probe that separates the models: $\chi^2 = 11.41$ for SSEE against 31.98 for ΛCDM at Planck (13 points); ΛCDM recovers 10.54 only by lowering Ω_m to 0.297. The supernova compilations share objects and the two CMB-lensing probes share sky, so the rows are not summed.

Table 3: χ^2 of SSEE and ΛCDM probe by probe, read from the analysis logs. Block A: background fixed in both models in advance (ΛCDM at the Planck 2018 best fit). Block B: ΛCDM fits cosmological parameters that SSEE does not have (column "fitted", SSEE / ΛCDM). $\Delta\chi^2 = \chi^2_{\rm SSEE} - \chi^2_{\Lambda\rm CDM}$. "$N$" is the number of degrees of freedom where the log records it. [a]Seven nuisance parameters minimised identically in both models. [b]TATT intrinsic alignments (the survey's baseline); with NLA the difference is $\Delta\chi^2 = +0.41$.

Probe	Type	N	fitted	$\chi^2_{\rm SSEE}$	$\chi^2_{\Lambda\rm CDM}$	$\Delta\chi^2$
A. Background fixed in both models						
Pantheon+	SN	1579	0 / 0	1386.62	1387.33	−0.70
DES-SN5YR	SN	1819	0 / 0	1629.44	1632.41	−2.97
Union3	SN	21	0 / 0	23.85	26.37	−2.52
Union3.1	SN	21	0 / 0	26.60	28.83	−2.24
ACT DR6	CMB lensing	10	0 / 0	14.01	14.20	−0.20
SPT-3G D1 KK[a]	CMB lensing	10	0 / 0	17.76	18.31	−0.54
DES Y3[b]	3×2pt	–	0 / 0	516.75	517.29	−0.54
DESI DR2 BAO	BAO	13	0 / 0	11.41	31.98	−20.58
B. ΛCDM fits parameters that SSEE does not have						
DESI DR2 BAO	BAO	13 / 11	0 / 2	11.41	10.54	0.87
Planck plik_lite	CMB	–	2 / 6	1003.59	1003.60	−0.01
Planck plik (full)	CMB	–	2 / 6	2768.45	2771.23	−2.78
KiDS-Legacy	shear	–	0 / 1	415.59	415.74	−0.14

4 CMB Angular Spectrum: CLASS Boltzmann Validation

4.1 CLASS implementation

We run the CLASS Boltzmann code [Lesgourgues, 2011] (v3.3, `src/p03_cmb/class_picos.py`) with the canonical SSEE background and a ΛCDM reference.

SSEE full-Ω_m (ω_m-direct canonical): $\omega_b = 0.022418$, $\omega_c = 0.119514$ and one massive neutrino at $\sum m_\nu = 0.06849$ eV (total $\Omega_{m,\rm CMB} = 0.308881$; OP-8 dissolved, no matter factor), $w_0 = -0.839950$, $w_a = -0.669975$, $n_s = 0.965558$, $h = H_0^{\rm glob}/100$. The full matter density is set directly as a CLASS background input.

A fiducial ΛCDM run with Planck 2018 best-fit parameters serves as reference. All runs use $\ell_{\rm max} = 2500$, lensing enabled.

4.2 Acoustic peaks with the single matter density

Table 4 presents the CMB acoustic peak positions and the RMS residual $\Delta C_\ell / C_\ell^{\Lambda\mathrm{CDM}}$ across $\ell = 30$–2500.

Table 4: CMB TT acoustic peaks and RMS residual versus ΛCDM. Peak positions identified from CLASS output $D_\ell = \ell(\ell+1)C_\ell/2\pi$.

Model	ℓ_1	ℓ_2	ℓ_3	RMS residual
ΛCDM (Planck 2018)	220	536	813	reference
SSEE ($\Omega_{m,\mathrm{CMB}} = 0.308881$)	**220**	**536**	**813**	**0.18%**

With $\Omega_{m,\mathrm{CMB}} = 0.308881$, the peak positions agree with ΛCDM to within $\Delta\ell \leq 1$ ($\ell = 220, 536, 813$) and the RMS residual is 0.18%, consistent with the $\chi^2_r(\mathrm{TT}) = 1.042$ reported in the CAMB-based Planck 2018 likelihood analysis of Paper 3 [Almeida Vallejo, 2026d]. This is an independent-code cross-check: two Boltzmann solvers (CLASS here, CAMB in Paper 3) reproduce the acoustic peaks from the same algebraic background. Earlier versions of this section added a third run that put the equation-of-state number $s_m = 1 + w_0 = 0.160050$ in the density slot and read its degraded peaks as evidence for a "two-Ω_m structure". s_m is not a density, so that run is not a version of the model; the comparison and the structure are withdrawn.

Figure 1 shows the TT power spectrum for SSEE and ΛCDM.

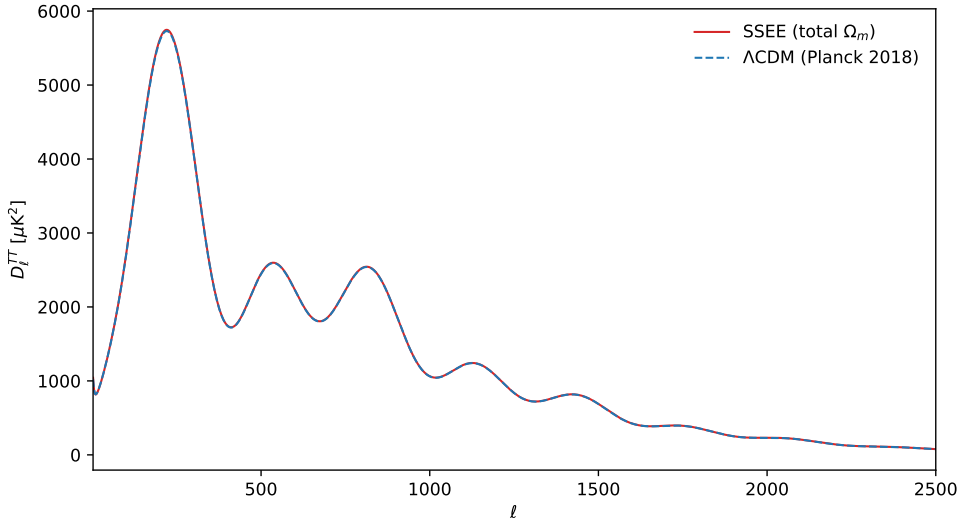


Figure 1: CMB TT power spectrum from CLASS. *Blue dashed*: ΛCDM Planck 2018. *Red solid*: SSEE with the full ω_m-direct canonical matter density $\Omega_{m,\mathrm{CMB}} = 0.308881$ ($w_0 = -0.839950$, $w_a = -0.669975$) — peaks at $\ell = 220, 536, 813$, agreeing with ΛCDM to within $\Delta\ell \leq 1$ (RMS = 0.18%).

4.3 Acoustic-scale mapping and r_d

The CAMB-based analysis of Paper 3 computes the SSEE drag horizon from the full ω_m-direct CMB density $\Omega_{m,\mathrm{CMB}} = 0.308881$ (OP-8 dissolved, no matter factor): the physical

drag horizon is $r_d = 147.17$ Mpc, 0.32σ from the Planck 2018 value $r_d = 147.09 \pm 0.26$ Mpc [Planck Collaboration, 2020] (TT,TE,EE+lowE+lensing, Table 2; no BAO prior). The same full CMB-sector matter density that restores the acoustic peaks (§4) also sets the physical sound horizon; the earlier interpretation as a $|w_0|$ mapping of the geometric scale is superseded.

5 Structure Growth, σ_8, and the $f\sigma_8$ Tension

5.1 Israel-Stewart causal perturbations: single-sector

Because $w_0 = -0.839950 < -1/3$, a naive k-essence interpretation yields a bare sound speed $c^2_{s,\mathrm{bare}} = w_0 \simeq -0.840$, implying catastrophic gradient instability on all sub-horizon scales. Paper 5 [Almeida Vallejo, 2026f] performs a causal Israel-Stewart (IS) analysis [Israel and Stewart, 1979]. The SSEE structure constants make the high-frequency IS sound speed vanish, $c^2_{s,\mathrm{eff}}(\omega\tau_\Pi \to \infty) = 0$, but that is *not* a stability result: with $c^2_{s,\mathrm{bare}} = w_0$ the growing mode is unstable at every k. The stability of the dark-energy sector is carried instead by the field of Paper 7, whose own sound speed is $c_s^2 = +0.021284$; with it no mode grows.

Integrating the linear growth equation from $z = 19$ to $z = 0$ with $\Omega_{m,\mathrm{cosm}} = 0.308881$ in the Poisson source:

$$\mathcal{G} = D_1^{\mathrm{SSEE}}/D_1^{\Lambda\mathrm{CDM}} = 1.0032, \qquad \sigma_8^{\mathrm{SSEE}} = 0.8136 \ \ (A_s \text{ held at Planck's value}). \tag{7}$$

The IS growth index $\gamma_{\mathrm{IS}} = 0.5504 \pm 0.0003$, consistent with $\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$ [Lahav et al., 1991], indicating that the SSEE perturbation growth rate tracks ΛCDM at the percent level despite the different background.

5.2 The growth amplitude, measured on raw shear data

The growth sector has a single matter density, $\Omega_m = 0.308881$, entering the background and the growth of structure alike, and it is tested against raw survey data rather than against compressed statistics whose reduction assumes a ΛCDM background.

Cosmic shear. Against the 357 unmasked *raw* $\xi_\pm$ points of KiDS-Legacy (Wright et al. 2025, arXiv:2503.19441), with the background fixed by the algebra and the amplitude free, the shear asks for $\log(10^{10}A_s) = 3.0544 \pm 0.0383$, -0.25σ from the 3.04483 that the model's own CMB fit supplies. Fixing A_s to that CMB value leaves the growth sector with *no* free cosmological parameter — only the eight survey nuisance parameters — and costs almost nothing: $\chi^2_{\mathrm{min}} = 415.59$ against 414.99 with A_s free ($\Delta\mathrm{BIC} = 5.28$ in favour of fixing it). The amplitude is then an output,

$$S_8 \equiv \sigma_8\sqrt{\Omega_m/0.3} = 0.8273, \qquad \text{measured } 0.8265 \pm 0.0176 \ (0.04\sigma), \tag{8}$$

with no posterior width of its own. KiDS-Legacy predates this analysis by sixteen months, and no temporal priority is claimed: the result is a consistency of A_s between the CMB and the growth epoch. The three χ^2 compared (fixed A_s, free A_s, Planck background) span only 0.74, so cosmic shear does not discriminate between the backgrounds; what separates them is the parameter count.

5.3 $f\sigma_8$

At the single-sector level with the Poisson source at $\Omega_{m,\mathrm{CMB}} = 0.308881$ (Paper 5), the growth factor is $\mathcal{G} = 1.0032$ and the mean $f\sigma_8$ tension across six RSD surveys is 0.70σ — statistically indistinguishable from ΛCDM (0.73σ). Against the *raw* BOSS DR12 redshift-space multipoles (144 points, one-loop LPT, Alcock–Paczynski applied to the model), and only after a ΛCDM control with its background free reproduces the published full-shape analysis of the same data, the algebraic background with every cosmological parameter fixed — amplitude included — reaches $\chi^2_{\mathrm{min}} = 117.10$, against 114.26 for ΛCDM with three more free parameters ($\Delta\chi^2 = +2.84$, $\Delta\mathrm{BIC} = -12.1$) and 115.34 for ΛCDM fixed at Planck, each with its own $\sum m_\nu$ (Paper 6).

This analysis operates at the linear-growth level for the growth index; non-linear effects (baryonic feedback beyond HMcode, halo bias) are not included.

6 EFT Stability Analysis

6.1 Bellini-Sawicki α-functions

Paper 7 [Almeida Vallejo, 2026h] writes the dark-energy sector as a k-essence scalar field with the two-term kinetic function $K(X) = c_1 X + c_2 X^2$, minimally coupled to gravity and to matter, with no potential. The ratio $u \equiv c_2 X/c_1 = -(M_v+T_r)/(3T_r+M_v) = -0.522735$ is fixed by the algebraic equation of state, and reproduces $w_\varphi = w_0 = -0.839950$ exactly. In the Bellini-Sawicki parametrisation [Bellini and Sawicki, 2014] this yields

$$\alpha_T = \alpha_M = \alpha_B = 0 \quad \text{(exact)}, \qquad \alpha_K = 3\,\Omega_{\mathrm{DE}}\,(5 - 3w_0) = 15.591336, \tag{9}$$

with the dark-energy *density* $\Omega_{\mathrm{DE}} = 1-\Omega_{m,\mathrm{CMB}} = 0.691119$. The vanishing of α_T, α_M and α_B implies: (i) gravitational waves propagate at c (GW170817 compatible at $|\alpha_T| < 10^{-15}$ [Abbott et al., 2017, of All-sky X-ray Image , MAXI]); (ii) no extra polarisation modes; (iii) standard light deflection.

6.2 Stability and what is observable

The solution requires $c_1 < 0$ and $c_2 > 0$: the field is a ghost condensate, stabilised at $X/X_{\mathrm{min}} = 1.045471$, where the no-ghost condition $K_X + 2XK_{XX} > 0$ holds. Its sound speed is fixed by the same algebra,

$$c_s^2 = \frac{M_v - T_r}{5M_v + 3T_r} = 0.021284 > 0, \tag{10}$$

so the sector is gradient-stable. An independent HI_CLASS integration reproduces c_s^2 to 0.000% and α_K to 0.011%, with the internal stability tests enabled (Paper 7). The kineticity itself is *not* observationally accessible: a factor 100 change moves σ_8 by 0.007% and the TT spectrum only at the lowest multipoles, below cosmic variance (Paper 7). The falsifiable content of this sector is therefore c_s^2 and w_0, not α_K.

The quantity 0.403302 that appears elsewhere in this journal is not a kineticity: it is $s_K = -3w_0(1 + w_0) = 3\,s_{\mathrm{DE}}\,s_m$, the fractional dilution rate of the dark-energy pressure per e-fold, built from the saturations $s_{\mathrm{DE}} = |w_0| = 0.839950$ and $s_m = 1 + w_0 = 0.160050$ (numbers of the equation of state, not densities). It enters f_{screen} (§7).

6.3 CPL phantom crossing and discriminating prediction

When the CPL EoS $w(z) = w_0 + w_a z/(1+z)$ is used in the EFTCAMB run, the phantom boundary $w = -1$ is crossed at $z = 0.314$. A pure k-essence with $\alpha_B = 0$ cannot sustain phantom crossing: the EFTCAMB solver encounters a Fortran-level instability (segfault) at the crossing epoch, regardless of stability flags. Earlier versions read this as evidence that completing the SSEE action in the phantom regime requires $\alpha_B \neq 0$, to supply a degree of freedom able to absorb the null-energy-condition violation. That reading is superseded. In the action of Paper 7 the field is minimally coupled and fixes w_0 and c_s^2 today, but it is not static: it rolls towards $w = -1$, with its own $w_a^\phi = +0.41$, and a single canonical field cannot cross $w = -1$. The algebraic $w(a)$ does cross it; what produces the crossing is open (OP-24). The conformal dark-sector coupling of earlier versions, which produced a crossing, is withdrawn.

For the action of Paper 7, $\alpha_B = \alpha_M = 0$ exactly, so a detection of $\alpha_B \neq 0$ by DESI Y3 or Euclid would falsify that action. What was a discriminating prediction conditional on a two-sector picture — an extension withdrawn on 2026-08-01 — is now a test of the action itself.

7 Extensions: Strong-Field Regime, Hubble Tension, and UV Completion

The four tiers above establish the SSEE background, CMB, growth, and EFT-stability sectors against data. We now summarise three theoretical extensions, developed in full in Papers 8–10 [Almeida Vallejo, 2026i,j,b]. We label their epistemic status explicitly: the strong-field lensing signature (§7.1) is a falsifiable prediction; the Hubble-tension screening fraction (§7.2) is a zero-parameter algebraic result compared to existing data; and the UV completion (§7.3) contains one quantity — $H_0^{\rm glob} = H_{\rm SH0ES}(1 - f_{\rm screen}) = 67.96214\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (residual $+4.2 \times 10^{-6}$ against the dimensionless target $3(\varphi+\pi)^2$, not a significance: the propagated uncertainty is ± 0.968) — whose UV term rests on Conjecture C.1 and is therefore presented as a consistency check, not an independent prediction.

7.1 Strong-field regime: lensing in the canonical limit

The action of Paper 7 is minimally coupled: it contains no disformal or conformal coupling between ϕ and dark matter, and photons follow the null geodesics of $g_{\mu\nu}$. Paper 8 was first written with a disformal coupling, whose null geodesic amplified the Einstein radius by $\sqrt{\rm AURA} \approx 1.9995$ (a MOND-like limit); that limit is kept there only as a historical contrast and makes no prediction of the model. *In the canonical limit* — a single cold matter sector with $\omega_c = {\rm KAL}_0\,\omega_b\,n_s$ (Paper 1, OP-8 closed) plus the Bellini–Sawicki structure $\alpha_B = \alpha_M = \alpha_T = 0$ of Paper 7, which gives $\mu - 1 = 0$ — the residual fifth-force is suppressed to $F_\phi/F_N \sim (H/k)^2 \sim 6 \times 10^{-13}$ at 10 kpc and the canonical observable lensing reduces to $\theta_E^{\rm SSEE} \approx \theta_E^{\rm GR-with-DM}$. Current SLACS/BELLS samples (mean total slope $\gamma = 2.078 \pm 0.027$, Auger et al. 2010; $M_E/M_{\rm dyn} \approx 1.21$ for SIS, Li et al. 2018) are consistent with the canonical limit. A $\sim \mathcal{M}$ matter-power suppression above a ϕ-DM free-streaming scale was previously listed here as a falsifiable prediction; it is retracted with the particle (§5.2).

7.2 The Hubble tension: an algebraic local screening fraction

The equation of state of Paper 1, with the structural constants of the algebra, fixes a late-time *local screening fraction*

$$f_{\rm scr}^{\rm IR} = \frac{s_K}{3\,\mathcal{M}} = \frac{\pi - \varphi}{(\varphi + \pi)^2} \approx 0.0673, \tag{11}$$

where $s_K = -3w_0(1 + w_0) = 0.403302$ is the pressure dilution rate of §6.2 and $\mathcal{M} = \mathrm{AURA}/2 = (3\varphi + \pi)/4$, a structural constant. The structural constant $\mathrm{AURA} = 2\mathcal{M}$ cancels exactly in the ratio, leaving a pure function of φ and π. The value is algebraic; its multiplicative action on the measured local rate is a postulate, since the disformal and separate-universe derivations of earlier versions are withdrawn (Paper 9), and its mechanism is open. The only directly measured expansion rate is the local one, $H_0^{\rm SH0ES} = 73.04 \pm 1.04$ [Riess et al., 2022]; Planck's is inferred within ΛCDM rather than measured, so SH0ES is the input and the global rate is the output,

$$H_0^{\rm glob,IR} = H_0^{\rm SH0ES}\,(1 - f_{\rm scr}^{\rm IR}) = 73.04 \times (1 - 0.06725) \approx 68.13\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{12}$$

which lies 0.17σ from the pure algebraic number $3(\varphi + \pi)^2 = 67.96214$ with zero free parameters. (With the model's ω_b, ω_c fixed, plik_lite places its own minimum at 67.87 ± 0.10, 0.90σ from $H_0^{\rm glob}$; Paper 3.) This is the infrared regime ($M \to \infty$), not the canonical value: with the complete screening fraction of the UV completion (§7.3) the single global rate is $H_0^{\rm glob} = 67.96214$. (The earlier MIRA anchor is superseded by the reframe.) The correction applies to local measurements of the rate; it does not modify the sound horizon r_d, the CMB peak positions, or the growth factor σ_8 established in the tiers above.

7.3 UV completion: the X^2/M^4 operator

The infrared screening uses only the equation of state. Paper 10 adds a kinetic function with a higher-derivative operator, $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$, and conjectures (Conjecture C.1) that the inflationary α-attractor parameter $\alpha = \varphi^4/3$ (Paper 1, Appendix A.5), which fixes the tensor-to-scalar ratio $r = 12\alpha/N_*^2 = 0.008131$, also sets the UV cutoff:

$$M^4 = 45\,\alpha^2\,\rho_c = 5\,\varphi^8\,\rho_c \approx 234.9\,\rho_c, \qquad M = 9.68\ \mathrm{meV}, \tag{13}$$

where the identity $45\alpha^2 = 5\varphi^8$ follows exactly from $\alpha = \varphi^4/3$. With this cutoff the full pressure-dilution rate (not the Bellini–Sawicki kineticity α_K) becomes $s_K^{\rm full} = 0.41691$ (+3.37% over the IR value). Applied to the measured SH0ES rate this gives the single global rate $H_0^{\rm glob} = 73.04 \times (1 - 0.069522) = 67.96214\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (residual $+4.2 \times 10^{-6}$ against $3(\varphi + \pi)^2$, with the unrounded $f = 0.069521611144$; propagated ± 0.968; the earlier MIRA-anchored value is superseded by the reframe).

We state the epistemic status of these figures plainly: the UV term rests on Conjecture C.1 and is *not independent of SH0ES*, which is the input of the cascade. The cascade does not measure the cutoff: varying M^4 from $0.2\times$ its value to infinity moves the output by at most $0.57\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, inside the propagated uncertainty. The IR value $H_0^{\rm glob,IR} = 68.13$ of §7.2 is the same cascade with the IR term alone, not a second rate. The construction does supply a cross-epoch test, but its decisive leg is r: the Hubble leg would vote only if LiteBIRD found $r < 0.0028$ (Paper 10).

8 Falsifiability and Prediction Register

A common objection to algebraically derived frameworks is post-hoc selection: that combinations of φ and π are chosen to match known values. The defence is *structural*, not chronological: the constants are drawn from a closed dictionary fixed by construction, with no optimisation over formula space except for ω_b (chosen among 7 expressions; OP-1). Table 5 classifies each derivation by epistemic status. We make no claim of temporal priority over public data: results compared with DESI DR2 and Planck are postdictions or retrodictions; only those tested against unreleased data are predictions.

Global falsification criterion. The SSEEframework would be falsified at the theory level if *any* of the following were established: (i) (w_0, w_a) is inconsistent with the DESI DR2 posterior at $> 3\sigma$; (ii) a future CLASS MCMC with Planck+DESI likelihoods requires $\Omega_{m,\mathrm{CMB}} \neq 0.308881$ at $> 3\sigma$; (iii) cosmic shear places S_8 more than 3σ from 0.8273 — the value predicted once A_s is *fixed* to what the CMB gives under the same background, leaving no free cosmological parameter and nowhere to absorb the discrepancy (superseding the weaker earlier form, 3σ from 0.7559 ± 0.0189 with A_s fitted to raw $\xi_\pm$ data and the SSEE background held fixed); (iv) $r \neq 0.00813$ as measured by LiteBIRD; (v) strong-lensing Einstein radii depart from GR with the model's cold matter at $> 3\sigma$ in HST/JWST surveys (the $\sqrt{\mathrm{AURA}}$ ratio belongs to the non-canonical disformal contrast of Paper 8 and is not a prediction).

Table 5: Predictive register: SSEE derivations by epistemic status. *Postdiction*: algebraic result fixed by the closed dictionary, then compared to data already public—no claim of temporal priority. *Retrodiction*: derived independently, then compared to existing data. *Prediction*: fixed now, tested against not-yet-released data.

Quantity (algebraic formula)	Value	Test / Instrument	Status
Background sector (algebraic dictionary)			
$w_0 = -T_r/M_v$	-0.839950	DESI DR2 CPL posterior	Postdiction
$w_a = -P_{\mathrm{sc}}/I_g$	-0.669975	DESI DR2 CPL posterior	Postdiction
$\mathcal{M} = (3\varphi + \pi)/4$	1.998924	$f_{\mathrm{scr}} = s_K/(3\mathcal{M})$ (Paper 9)	Postdiction
Retrodictions (derived independently, compared to existing data)			
$H_0^{\mathrm{glob}} = H_{\mathrm{SH0ES}}(1 - f_{\mathrm{screen}})$, target $3(\varphi + \pi)^2$	67.96214 km/s/Mpc	Planck 2018	Retrodiction
$n_s = 1 - \varphi^{-7}$	0.965558	Planck 2018	Retrodiction
$\Omega_{m,\mathrm{CMB}} = \omega_m/h^2$ (ω_m-direct)	0.308881	Planck 2018	Retrodiction
r_d (CAMB, ω_m-direct)	147.17 Mpc	Planck 2018 r_d (0.32σ)	Retrodiction
γ_{IS}	0.5504 ($\gamma_{\Lambda\mathrm{CDM}} \simeq 0.55$)	RSD surveys	Retrodiction
$\alpha_T = \alpha_M = \alpha_B = 0$	exact	GW170817 $\lvert\alpha_T\rvert < 10^{-15}$	Retrodiction
S_8 (KiDS-Legacy, A_s fixed by the CMB)	0.8273 predicted	measured 0.8265 ± 0.0176 (0.04σ)	Retrodiction (nothing fitted)
earlier: KiDS-1000, A_s fitted (superseded)	0.7559 ± 0.0189	ΛCDM control: 0.7571 ± 0.0194	Retrodiction
σ_8 (A_s fixed to Planck; ceiling)	0.8149	convention, not a prediction	—
$H_0^{\mathrm{glob,IR}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{scr}}^{\mathrm{IR}})$	68.13 km/s/Mpc	$3(\varphi + \pi)^2 = 67.96214$ (0.17σ; IR only, partial)	Retrodiction
$H_0^{\mathrm{glob}} = H_0^{\mathrm{SH0ES}}(1 - f_{\mathrm{scr}}^{\mathrm{full}})$	67.962142 km/s/Mpc	$3(\varphi + \pi)^2$ (residual $+4.2 \times 10^{-6}$)	Retrodiction
Predictions (not yet observationally accessible)			
$c_s^2 = (M_v - T_r)/(5M_v + 3T_r)$	0.021284	dark-energy clustering (Euclid, LSST)	Prediction
$r = \varphi^{-10}$	0.008131	LiteBIRD	Prediction
$\theta_E^{\mathrm{SSEE}}/\theta_E^{\mathrm{GR}}\lvert_{\mathrm{alt}} = \sqrt{\mathrm{AURA}}$	1.9995 (alt. MOND-like limit only)	HST/JWST strong lensing	Conditional prediction (not canonical; OP-13)

9 Discussion and Outstanding Challenges

The consolidated results show that SSEE is consistent with four independent observational tests without adjusting any algebraic parameter. The most significant open challenges are:

(i) ω_m-direct residual dependencies. The CLASS run confirms that the ω_m-direct density $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$ (built from $\omega_b + \omega_c + \omega_\nu$, no matter factor; OP-8 dissolved) reproduces the CMB peaks, and it is *not* a free knob: $\omega_c = \mathrm{KAL}_0 \cdot \omega_b \cdot n_s$ is a forward algebraic identity (Paper 1). The honest residual is that $\Omega_{m,\mathrm{CMB}}$ now rests on ω_b (OP-1, partially resolved: the algebraic form is fixed, the BBN chain is not) and this ω_c identity being correct—a dependency chain, not an undetermined parameter.

(ii) S_8 at the non-linear level. The growth amplitude is fixed by the CMB and predicts $S_8 = 0.8273$ against raw KiDS-Legacy (§5.2). The open item is a precision question: a full N-body treatment of baryonic feedback (deferred for computational cost) is required to confirm the linear-theory S_8 at the non-linear level.

(iii) Quantum field theory completion. The coincidence between the measured cascade H_0^{glob} and the pure number $3(\varphi+\pi)^2$ is a comparison, not a derivation: deriving H_0 from first-principles quantum field theory remains open, for this and every model. Paper 10 supplies a UV completion at the EFT level — the higher-derivative operator X^2/M^4 with cutoff $M = 9.68$ meV (Conjecture C.1) — but a derivation of the full $P(X,\phi)$ Lagrangian from φ and π alone remains open. Under the ω_m-direct reframe (OP-8 dissolved) the CMB matter density is the standard $\Omega_{m,\mathrm{CMB}} = \omega_m/h^2 = 0.308881$, with no matter factor; the earlier MIRA-mapped and geometric $(\pi-\varphi)/(\pi+\varphi)$ forms are superseded. The residual open problem is now only the UV origin of the algebraic $\omega_c = \mathrm{KAL}_0\,\omega_b\,n_s$ identity.

10 Conclusions

We have presented a consolidated validation of the SSEEframework across four observational tiers.

1. **Background sector.** The algebraic prediction $(w_0, w_a) = (-0.839950, -0.669975)$, fixed by construction (not fitted), lies within 0.24σ of the official DESI DR2 w_0w_aCDM posterior (Pantheon+). A Bayesian MCMC yields $H_0 = 67.82 \pm 0.41$ km s^{-1} Mpc^{-1} under H_0^{glob} as prior (0.34σ from it), and $\Delta\mathrm{BIC} = -10.90$ favouring SSEE over ΛCDM with nothing free (-5.81 in the conservative $k = 2$ version). A complementary full-likelihood multi-probe MCMC (w_0, w_a free, H_0 free under the uncompressed CMB likelihood) recovers the CMB combinations $\omega_b, \omega_c, r_{\mathrm{drag}}$ to $\leq 0.2\sigma$ and places SSEE at 1.31σ in the w_0–w_a plane, where ΛCDM is excluded at 3.22σ.

2. **CMB Boltzmann.** With the single ω_m-direct matter density ($\Omega_{m,\mathrm{CMB}} = 0.308881$), CLASS places the acoustic peaks where ΛCDM does. With the full ω_m-direct density, peak positions agree with ΛCDM to within $\Delta\ell \leq 1$ (RMS = 0.18%).

3. **Structure growth.** The linear perturbation analysis ($\Omega_{m,\mathrm{CMB}} = 0.308881$ in Poisson) gives $\mathcal{G} = 1.0032$ and, with A_s at Planck's value, $\sigma_8 = 0.8136$, with mean $f\sigma_8$ tension 0.70σ (Paper 5) — statistically identical to ΛCDM (0.73σ). In the lensing amplitude, with A_s fixed to the value the model's own CMB fit supplies and no free cosmological parameter, a single matter sector predicts $S_8 = 0.8273$ against 0.8265 ± 0.0176 from

raw KiDS-Legacy $\xi_\pm$ (Paper 6). Against raw BOSS DR12 multipoles the algebraic background, with nothing fitted, gives $\chi^2_{\rm min} = 117.10$ against 114.26 for the published ΛCDM with three more free parameters (ΔBIC $= -12.1$).

4. **EFT stability.** The two-term k-essence of Paper 7 is a ghost condensate, ghost-free and gradient-stable with $c_s^2 = 0.021284$; $\alpha_T = \alpha_M = \alpha_B = 0$ exactly (GW170817 compatible) and $\alpha_K = 15.591336$, which is not observationally accessible. A detection of $\alpha_B \neq 0$ would falsify the sector. The same action fixes w_0 but does not produce w_a (OP-24).

5. **Theoretical extensions.** In the strong-field regime the canonical lensing is that of GR with the model's cold matter ($\mu - 1 = 0$, Paper 8). An algebraic late-time screening fraction $f^{\rm IR}_{\rm scr} = (\pi - \varphi)/(\varphi + \pi)^2 = 0.0673$ takes, under the screening postulate of Paper 9, the measured SH0ES rate as input and returns $H_0^{\rm glob,IR} = H_0^{\rm SH0ES}(1 - f^{\rm IR}_{\rm scr}) = 68.13\ {\rm km\,s^{-1}\,Mpc^{-1}}$ with the IR term alone, 0.17σ from the pure number $3(\varphi + \pi)^2 = 67.96214$, with zero free parameters (Paper 9). A UV completion $K(X) = X/{\rm KAL}_0 + X^2/M^4$ whose cutoff $M = 9.68$ meV is conjectured to be set by the inflationary α-attractor (Conjecture C.1) completes the fraction; the single global rate $H_0^{\rm glob} = 67.96214\ {\rm km\,s^{-1}\,Mpc^{-1}}$ (residual $+4.2 \times 10^{-6}$ against $3(\varphi + \pi)^2$, propagated ± 0.968) is a consistency check, not an independent prediction, and does not measure M (Paper 10).

The framework generates four near-term falsifiable predictions, under the stated assumptions (a fifth, a ϕ-DM free-streaming scale in $P(k)$, is withdrawn with the particle; §5.2): the dark-energy sound speed $c_s^2 = 0.021284$, $\alpha_B = 0$ (a detection would falsify the action of Paper 7), $r = 0.008131$ (LiteBIRD), and $S_8 = 0.8273$ with no free cosmological parameter (Paper 6). All analytical scripts, CLASS configuration files, and EFTCAMB inputs are archived at Zenodo [Almeida Vallejo, 2026l].

A Complete Algebraic Parameter Table

Acknowledgments

The author acknowledges the use of the cosmological codes CLASS [Blas et al., 2011], EFTCAMB [Hu et al., 2014], the affine-invariant ensemble sampler EMCEE [Foreman-Mackey et al., 2013], which ran the DESI DR2 BAO chains of the model comparison, and the Markov Chain Monte Carlo framework COBAYA [Torrado and Lewis, 2021], which ran the CMB, weak-lensing and BOSS chains. The author also acknowledges the use of the GETDIST package [Lewis, 2019] for the analysis and plotting of posterior distributions.

Data and Code Availability

The SSEE framework is fully transparent and reproducible. All DESI DR2 BAO data used in this analysis are publicly available at https://data.desi.lbl.gov/, and the Planck 2018 cosmological parameter chains at the Planck Legacy Archive (https://pla.esac.esa.int/). The complete code repository — including the CLASS and EFTCAMB configuration files, the COBAYA MCMC chains, the explicit record of Open Theoretical Problems (OP), and the cosmological validation ledgers — is publicly available at

Table 6: Complete set of SSEEalgebraic parameters. All values are derived from $\varphi = (1+\sqrt{5})/2$ and π; no fitting was performed. Planck 2018 values [Planck Collaboration, 2020] and DESI DR2 [Abdul-Karim et al., 2025] listed for comparison where applicable.

Parameter	Formula	SSEE value	Observed
w_0	$-T_r/M_v = -(3\varphi+\pi)/[2(\varphi+\pi)]$	-0.839950	-0.838 ± 0.055 (DESI DR2+CMB+Pantheon+, chain moments)
w_a	$-P_{\rm sc}/I_g$	-0.669975	-0.617 ± 0.208 (DESI DR2+CMB+Pantheon+, chain moments)
$H_0^{\rm glob}$	$H_{\rm SH0ES}(1-f_{\rm screen})$, target $3(\varphi+\pi)^2$	67.96214 km/s/Mpc	67.36 ± 0.54 (Planck)
n_s	$1-\varphi^{-7}$	0.965558	0.9649 ± 0.0042 (Planck)
$\Omega_{m,\rm CMB}$	ω_m/h^2 (ω_m-direct)	0.308881	0.3153 ± 0.0073 (Planck)
$\Omega_{\rm DE}$	$1-\Omega_{m,\rm CMB}$ (density)	0.691119	0.6847 (ΛCDM)
$s_{\rm DE}$	$T_r/M_v = \lvert w_0\rvert$ (saturation, not a density)	0.839950	—
s_m	$1-s_{\rm DE} = 1+w_0$ (saturation, not a density)	0.160050	—
$\mathcal{M}$	$(3\varphi+\pi)/4$ (Mirror Ratio; enters $f_{\rm scr}$)	1.998924	—
KAL_0	$\beta+\pi = (\varphi+3\pi)/2$	5.521406	—
α_K	$3\,\Omega_{\rm DE}(5-3w_0)$	15.591336	not accessible
s_K	$-3w_0(1+w_0)$	0.403302	— (enters $f_{\rm scr}$)
c_s^2	$(M_v-T_r)/(5M_v+3T_r)$	0.021284	—
$\alpha_T=\alpha_M=\alpha_B$	minimal coupling	0	$\lvert\alpha_T\rvert < 10^{-15}$ (GW170817)
S_8	KiDS-Legacy, A_s fixed by the CMB	0.8273 predicted	0.8265 ± 0.0176 (KiDS-Legacy)
S_8	earlier: KiDS-1000, A_s fitted (superseded)	0.7559 ± 0.0189	0.759 ± 0.024 (KiDS-1000)
r	φ^{-10}	0.008131	< 0.036 [BICEP/Keck Collaboration, 2021]
$f_{\rm scr}^{\rm IR}$	$(\pi-\varphi)/(\varphi+\pi)^2$	0.0673	—
$H_0^{\rm glob}$	$H_0^{\rm SH0ES}(1-f_{\rm scr}^{\rm full})$	67.962142 km/s/Mpc	$3(\varphi+\pi)^2$, residual $+4.2\times10^{-6}$
M (UV cutoff)	$(5\varphi^8\rho_c)^{1/4}$ (Conj. C.1)	9.68 meV	—

the GitHub repository https://github.com/mikealmeida1721/SSEE-Cosmologia and archived at Zenodo: https://doi.org/10.5281/zenodo.19654118.

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